

GARRICK TSE Edmonton, AB 825-440-5178 g.tse8888@gmail.com github.com/Garrrrrrrr GarrickTse.com LinkedIn	
TECHNICAL SKILLS	
Languages: Python, C, Java, JavaScript, TypeScript, SQL, HTML/CSS, PHP Databases: PostgreSQL, Microsoft SQL Server, MongoDB, Firebase Firestore Web Technologies: React, Next.js, Node.js, Express, REST APIs, Tailwind CSS Frameworks & Libraries: Pandas, NumPy, Matplotlib, BeautifulSoup, Tkinter Tools & Platforms: Git/GitHub, Supabase, Vercel, AWS, Android Studio, Windows, Linux, Jupyter Notebook Core Skills: Requirements Gathering, Problem-Solving, Teamwork, Time Management, Leadership	
EDUCATION	
Bachelor of Science, Computing Science – Specialization University of Alberta Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, AI Search and Planning	Sep. 2023 – Present Edmonton, AB
PROJECTS	
Gateway Training Tool TypeScript, React, Tailwind CSS, PostgreSQL, Supabase • Architecting and developing a full-stack web application supporting Gateway training operations across Canada. • Built a React + TypeScript frontend enabling coordinators to manage classes and trainer assignments, trainers to create drills and grade students, and students to track performance metrics and drill times. • Designed a PostgreSQL schema modeling Coordinator → Trainer → Student workflows with Supabase authentication, row-level security, and file storage for reports and PDFs. • Deployed the frontend with Vercel and backend infrastructure with Supabase.	In Progress
CountLab.ca TypeScript, Next.js, Tailwind CSS, Supabase, Python • Built and deployed a full-stack blackjack card-counting trainer with Supabase authentication and per-user progress synchronization. • Designed a bankroll analytics dashboard with Kelly-criterion bet ramps, risk-of-ruin calculations, and EV/variance visualizations driven by simulation data. • Engineered a Python Monte Carlo simulation engine generating 46.7B+ blackjack rounds across 100M shoes to estimate EV and risk coefficients. • Developed an exact game-theoretic solver for a casino poker variant using a Numba-compiled backward-induction kernel evaluating 18B+ terminal states.	
Prominent Person Stock Tracker Python, Pandas, BeautifulSoup, Tkinter • Built a desktop stock-tracking application visualizing publicly disclosed trades from executives and politicians across multiple data sources. • Scraped and aggregated data using BeautifulSoup, processing millions of data points through a structured Pandas pipeline. • Developed interactive filtering by individual and ticker, plus downloadable charts and CSV exports.	
Android Social Media Mood App Java, Firebase, Android Studio • Developed a scalable Android application in an Agile team using GitHub workflows. • Optimized database queries and asynchronous handling, improving efficiency by 74%. • Designed a distributed Firebase backend with client-side caching.	
Distributed Social Networking Platform Python, Django, PostgreSQL, REST APIs, Heroku • Collaborated in a 5-person team to design and test a federated social networking platform and validate cross-node interactions. • Developed API test cases covering authors, posts, comments, likes, follow requests, authentication rules, edge cases, and pagination. • Investigated defects by reproducing bugs, analyzing root causes, and verifying fixes before deployment.	
Ray Tracing Engine C • Optimized 3D rendering algorithms, improving render time by 100%. • Implemented advanced lighting and shading models.	
EXPERIENCE	
Dealer + Training Software Developer Gateway Casinos and Entertainment • Built Gateway Training Tool, a full-stack platform supporting trainer assignments, student progress tracking, drill creation, and performance reporting. • Calculated and distributed winnings with 100% accuracy in a fast-paced casino environment. • Managed hiring processes for 40+ candidates and trained new dealers. • Supervised trainees to ensure operational compliance and service quality.	Sep. 2024 – Present Edmonton, AB